

Implement Threat Intelligence Principles

Progress Report — IoC Analysis Complete, OpenCTI Implementation Pending

Project Name	Healthcare IAM Operations Simulation
Organization	Horizon Valley Medical Center (HVMC)
Rubric	Implement Threat Intelligence Principles
Document Version	0.5 (partial — see Status below)
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1. Rubric Status

Rubric Requirement	Status	Notes
Analysis of 2 IoCs: detection methods, how each indicates a threat	Complete	See Section 2
OpenCTI implementation (Docker/system), 2 connectors configured, platform + connector documentation, usage demo	Pending	Not yet implemented — see Section 3 for the scoped, ready-to-execute plan

This report is submitted at 50% completion by design: the IoC analysis component is genuine, finished work; the OpenCTI platform component has not been built yet, so it is documented here as a concrete implementation plan rather than a false claim of a working system. Marking this rubric “Complete” requires the OpenCTI platform to actually be running with two functioning connectors — that evidence does not yet exist, and this report does not represent otherwise.

2. IoC Analysis (Complete)

Both indicators are drawn from the JOMANGY/INJ3CTOR3 sample already analyzed under the Identify and Analyze Cyber Threats rubric (Malware_Diagnosis_Report_JOMANGY_679515b3.docx), chosen deliberately as two different IoC types so the analysis demonstrates layered detection rather than two examples of the same technique.

2.1 IoC 1 — File Hash (SHA-256)

Indicator	679515b3d6642a102f3c8bb69c0884912bc700d2a575e2595752e8b446e39c00
Type	File-based IoC (cryptographic hash)
Detection Method	Multi-engine scanning via VirusTotal: the hash was queried against 60+ independent antivirus/security vendor engines, with 22 of 60 flagging the file as malicious. VirusTotal's automated Code Insights analysis additionally corroborated this by identifying unauthorized administrative/system-level modification behavior targeting FreePBX/Asterisk VoIP environments.
How It Indicates a Threat	A SHA-256 hash uniquely fingerprints one exact file — any single-byte change produces a completely different hash. Independent detection by 22 separate vendors provides multi-source corroboration that this specific artifact is malicious, reducing the chance of a single-engine false positive. This makes hash-based IoCs high-fidelity (a match is a near-certain confirmation) but narrow in scope: they only catch this exact file, and are trivially evaded if the attacker repacks or modifies the payload.

2.2 IoC 2 — Network Indicator (Command-and-Control IP)

Indicator	160.119.69.4
Type	Network-based IoC (IP address)
Detection Method	Static/behavioral analysis of the JOMANGY dropper script: manual code review and string extraction of the sample revealed a hardcoded curl command retrieving a second-stage payload from http://160.119.69.4/x. This was identified through source-level analysis of the malware itself rather than automated AV signature matching.
How It Indicates a Threat	This IP is the malware's actual payload-hosting/command-and-control infrastructure. Unlike a file hash, a network IoC can catch the same attacker infrastructure even across variants with different hashes — if JOMANGY is repacked into a new file, this IP may still appear in the new sample's traffic. This makes network IoCs useful for infrastructure-level detection (firewall blocklisting, DNS sinkholing, outbound traffic monitoring) independent of what the endpoint file looks like, though they carry more false-positive risk than file hashes if the IP is later reassigned or hosts unrelated services.

2.3 Why These Two IoC Types Together

The file hash and the network IP were chosen as a complementary pair rather than two examples of the same detection approach: the hash gives high-confidence identification of this specific artifact wherever it appears (email attachment, endpoint scan, download), while the C2 IP extends detection to the malware's infrastructure even if the file itself changes. A real detection program layers both — relying on either alone leaves a gap the other closes.

3. OpenCTI Platform Implementation (Pending)

This component has not been built as of this report date. The plan below is fully scoped and ready to execute; this section will be replaced with real screenshots, connector status, and usage evidence once implementation is complete.

3.1 Planned Deployment

Platform	OpenCTI, via the official Docker Compose stack (Platform, Elasticsearch, Redis, RabbitMQ, MinIO, Worker)
Host	HVMC-KALI, resized to 6GB RAM / 40GB disk with Elasticsearch heap capped at 1GB to fit the available hardware
Connector 1	MITRE ATT&CK connector — imports the same ATT&CK framework data already used for the Orangeworm/Kwampirs mapping in the Cyber Threats rubric
Connector 2	MalwareBazaar (abuse.ch) connector — the same source the JOMANGY sample itself came from
Usage Demonstration (planned)	Create the two IoCs above as Observables, promote each to an Indicator, link both to a Malware entity for JOMANGY, and capture the resulting relationship graph

3.2 Remaining Steps

- Resize HVMC-KALI (6GB RAM / 40GB disk) and grow the partition/filesystem.
- Install Docker and Docker Compose.
- Clone the OpenCTI docker repository and configure .env (admin credentials, UUID tokens, capped Elasticsearch memory).
- Launch the stack and confirm all containers report healthy; log into the web UI.
- Enable and verify the MITRE ATT&CK and MalwareBazaar connectors.
- Create and link the two IoCs from Section 2; capture screenshots of the platform, connector status, and indicator relationships.

4. Conclusion

The IoC analysis requirement is fully satisfied in Section 2, with two distinct, complementary indicator types genuinely analyzed and tied to existing project findings. The OpenCTI platform requirement remains outstanding; Section 3 documents a specific, executable plan rather than claiming functionality that does not yet exist. This report will be updated to a completed final version once the platform is deployed and evidence is captured.

5. References / Sources

- Malware_Diagnosis_Report_JOMANGY_679515b3.docx — source of both IoCs analyzed in Section 2.
- VirusTotal file report — SHA-256 679515b3d6642a102f3c8bb69c0884912bc700d2a575e2595752e8b446e39c00.
- OpenCTI Platform documentation — docs.opencti.io (Docker deployment, connector configuration).

Prepared for: Sabitha — Healthcare IAM Operations Home Lab | Cyber Threats and Vulnerabilities rubric — Implement Threat Intelligence Principles (progress report, not final)